

Reflection Mirror Map

Unit 9: Treasure Map Navigator · Lesson 9-7 · EduWonderLab

UNIT 9 · LESSON 9-7 · TEACHER NOTES

Standard 6.NS.6b

FORMAT Reflect + check game

TIME 25–30 min

GROUPING Pairs

TOPIC Reflect Points Across Axes

STANDARD 6.NS.6b

PREP LEVEL Light — print coordinate grids and reflection cards

Learning goal *I can reflect a point across the x-axis or y-axis and describe how the signs change.*

Teacher setup

- Print one set per pair. Students plot a point, then reflect it across the named axis.
- They identify how the coordinate signs changed.
- Partners check with a “mirror rule” card.

Materials

Coordinate grids, Reflection cards, Colored pencils, Pencil.

Differentiation & language support

- Mirror rule: across the x-axis, y changes sign; across the y-axis, x changes sign (SPED).
- Sentence stem: “Across the ___-axis, ___ changed and ___ stayed the same.”
- ESOL: use a small mirror or fold the paper to model reflection.

Key vocabulary (English / Español):

Term	Español	What it means
reflect	reflejar	to flip across a line
axis	eje	the x or y line of the grid

Quick teacher check: *Check Card 8 — identifying the axis from the change.*

Reflection Mirror Map

Unit 9: Treasure Map Navigator · Lesson 9-7 · EduWonderLab

Name: _____

Date: _____

Class: _____

Your goal: *I can reflect a point across the x-axis or y-axis and describe how the signs change.*

What you need

Coordinate grids, Reflection cards, Colored pencils.

How to play

1. Plot the point on the card.
2. Reflect it across the named axis.
3. Write the new ordered pair.
4. Describe which coordinate changed sign.

Reflection Cards

Across the x-axis the y-sign flips; across the y-axis the x-sign flips.

1. Reflect (3, 4) across the x-axis.	2. Reflect (3, 4) across the y-axis.
3. Reflect (-2, 5) across the x-axis.	4. Reflect (-2, 5) across the y-axis.
5. Reflect (6, -1) across the x-axis.	6. Reflect (-4, -3) across the y-axis.
7. Reflect (0, 5) across the x-axis.	8. (5, 2) was reflected to (5, -2). Which axis?

Record your work (new point + what changed):

Explain your thinking

Explain why reflecting across the x-axis changes only the y-coordinate.

Sentence frame: *Across the ___-axis, ___ changed and ___ stayed the same.*

★ **Challenge:** Reflect $(3, 4)$ across the x -axis, then across the y -axis. Where does it land? $(-3, -4)$

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 9 • LESSON 9-7 • Reflection Mirror Map

Standard 6.NS.6b

Answers

1. Reflect (3, 4) across the x-axis. Answer: (3, -4)	2. Reflect (3, 4) across the y-axis. Answer: (-3, 4)
3. Reflect (-2, 5) across the x-axis. Answer: (-2, -5)	4. Reflect (-2, 5) across the y-axis. Answer: (2, 5)
5. Reflect (6, -1) across the x-axis. Answer: (6, 1)	6. Reflect (-4, -3) across the y-axis. Answer: (4, -3)
7. Reflect (0, 5) across the x-axis. Answer: (0, -5)	8. (5, 2) was reflected to (5, -2). Which axis? Answer: The x-axis

Teaching notes & look-fors

- Across the x-axis: $(x, y) \rightarrow (x, -y)$. Across the y-axis: $(x, y) \rightarrow (-x, y)$.
- Challenge: two reflections land the point in the opposite quadrant.